



JORGE LEONARDO LORETO

AI Data Scientist · Econometrician · Financial Analyst (FMVA®)

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PROFESSIONAL SUMMARY

Economist and Data Scientist with 10+ years of experience transforming complex commercial and financial data into strategic business impact. Specialized in **time-series forecasting**, **econometric causal inference**, **large-scale retail modeling (85+ stores)**, and **autonomous AI agent engineering**. Strong corporate finance background (FMVA® · CFI) and cloud data pipeline architecture (PostgreSQL, BigQuery, Python, Docker). Trilingual professional (English, Spanish, French).

TECHNICAL SKILLS

Languages & Core Data: Python (Pandas, NumPy, SciPy, Statsmodels, Scikit-learn, PyTorch), SQL (SQL Server, BigQuery, PostgreSQL), DAX, Power Query, Bash.

Statistical Modeling & Machine Learning: Hierarchical Forecasting (Prophet, LSTM), Causal Inference (DiD, Synthetic Control, CausalImpact), Clustering (K-Means, Agglomerative Hierarchical, PCA), Monte Carlo Simulations, Iterative Proportional Fitting (IPF).

Artificial Intelligence & Agents: LLM API Integration (Anthropic Claude, OpenAI), Whisper STT, Multi-Tool Autonomous Agents (n8n self-hosted, Traefik, Docker), MCP Architecture, Advanced Prompt Engineering.

BI, Web Apps & Cloud: Power BI, Looker, Tableau, Streamlit, Plotly, Supabase, Google Cloud Platform (BigQuery), SAP BW, Docker Compose, Git / GitHub Actions.

PROFESSIONAL EXPERIENCE

- Head of Business Intelligence and Management Control**
Grupo Ramos · Sirena / Sirena Market / Aprezio

Nov 2025 — Present
Santo Domingo, DR

 - **Led** a multidisciplinary BI and management control analyst team overseeing commercial, logistics, and inventory execution across the country's leading retail network (**85+ stores in 17 provinces**).
 - **Architected** the analytical roadmap and defined executive KPIs reported to the Executive Committee and C-Level (Sales, Commercial Margin, FEG% On-Shelf Availability, Fill Rate, and Inventory Turnover).
 - **Engineered** integrated financial projection models for Sales, Margin, and EBITDA in Python and SQL Server, connecting SAP BW and BigQuery pipelines.
 - **Directed** as Business Lead the strategic cloud migration POC for **BigQuery + Looker**, unifying corporate data modeling with analytics governance.
- Data Scientist — Financial Control**
Grupo Ramos · Control Financiero Corporativo

Oct 2024 — Nov 2025
Santo Domingo, DR

 - **Built** the end-to-end **FY2026 Budget Distribution** pipeline across 5 hierarchical levels (Store → Family → Subfamily → Item) using *Iterative Proportional Fitting*, historical EWMA, and mirror-store logic with exact reconcile tolerance ($\leq$ RD\$ 1).
 - **Conducted** econometric causal inference using **Difference-in-Differences (DiD)** and **Synthetic Control** to evaluate campaign execution impact, optimizing SQL queries from **20M down to 800K rows**.
 - **Implemented** unsupervised clustering algorithms (K-Means + PCA) for structural and financial store segmentation across 85+ retail stores with ARI validation.
 - **Modeled** demand causality and product substitution in out-of-stock events (FEG%) using *Granger Causality* to protect high-margin core SKUs.
 - **Automated** daily extraction and transformation pipelines (**SAP BW → SQL Server → Power BI**) using Python, eliminating manual processing risks.

- **Managed** import/export financial and commercial operations, international cost structuring, customs compliance, and supply contract negotiations.
- **Analyzed** price trends and international freight volatility using quantitative econometric models applied to key vendor negotiations.

FEATURED AI & DATA ENGINEERING PROJECTS

RepubliCaraquistApp — Plataforma de Sports Analytics & Sabermetría

Python · Streamlit · Supabase (PostgreSQL) · MLB Stats API · Plotly

2024 — Presente

[Ver App en Vivo](#)

- Engineered a live sabermetrics production web app with automated play-by-play data ingestion pipeline from MLB Stats API to Supabase PostgreSQL.
- Architected a **24-state stochastic Win Expectancy engine (Tango RE24)** calculating real-time *Leverage Index (LI)* and *Clutch* metrics per play and inning.
- Implemented a dynamic **ELO Rating system** with home advantage (+35 pts) and **Monte Carlo simulations (5,000 iterations)** projecting regular season standing distributions and LVBP playoff probabilities.

Autonomous AI Personal Assistant

n8n (Self-Hosted) · Docker · PostgreSQL · OpenAI / Whisper · Telegram Bot

2025 — Presente

Self-Hosted Infrastructure

- Deployed an autonomous conversational agent on Hostinger KVM2 VPS with persistent memory (PostgreSQL) and multi-tool calling API integrations (Google Calendar, Gmail, Weather).
- Integrated Whisper voice note transcription and background reminder dispatcher workflows containerized on Docker + Traefik.

EDUCATION

Master of Science in Data Science applied to Business Intelligence · UNINI, Mexico

2025 — Present

Specialization in Money and Financial Institutions · Central University of Venezuela (UCV)

2015 — 2017

Bachelor of Science in Economics · Santa Maria University (USM), Caracas, Venezuela

2008 — 2014

PROFESSIONAL CERTIFICATIONS

AI Engineer for Data Scientists Associate · DataCamp

2026

Financial Modeling & Valuation Analyst (FMVA®) · Corporate Finance Institute (CFI)

2026

Data Scientist Associate · DataCamp

2025

Data Analyst Associate · DataCamp

2024

LANGUAGES

Spanish: Native | English: Full Professional (C1) | French: Professional / Intermediate (B2)

Trilingual